

Pre-Session Manual for Capacity Building Sessions

Thursday 21st August, 2025

This manual has been prepared to help you in preparation for the upcoming training sessions. We are keeping it simple – to ensure that nothing is lost in translation, and everyone is able to have all they need to make learning easy – so please do not be offended by the simplicity of this manual. If we all get this right, then the classes will flow smoothly.

The manual contains:

1. How to download and install R and RStudio and its subsequent packages for the sessions
2. How to download and install QGIS Software.
3. How to install KoboCollect App
4. How to join the morning and evening sessions
5. How to join the evening breakout sessions
6. How to use the Google Classroom platform.

1. Download 'R' and its interface 'RStudio'

a. Download and Installation of R

R is a statistical computing language, and it will be predominantly be used in this course.

To download R, you go to:

<https://cran.r-project.org/>

or use the link:

<https://posit.co/download/rstudio-desktop/>

It should look like this and as most of you are on windows, this one encircled is what you will download



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#) (Debian, Fedora/Redhat, Ubuntu)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2025-06-13, Great Square Root) [R-4.5.1.tar.gz](#), read [what's new](#) in the latest version.
- The CRAN directory [src/base-prerelease](#) contains R alpha, beta, and rc releases as daily snapshots in time periods before a planned release.
- Between releases, the same directory [src/base-prerelease](#) contains snapshots of current patched and development versions. Please read about [new features](#) and [bug fixes](#) before filing corresponding feature requests or bug reports.

Of course, if you have a Mac or Linux, kindly ensure you use the correct link.

Once you have downloaded, install the executable file by double-clicking on the file. Then follow the steps as instructed by the prompts to install the software.

b. Download and Installation of RStudio

Next you will download **RStudio** which is an integrated development environment for **R**. There are other similar software which include Vim, Eclipse, among others, but these will not be covered during this course. To download **RStudio**, follow this link:

<https://www.rstudio.com/products/rstudio/download/>

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 281.24 MB | SHA-256: 3A553330 | Version: 2025.05.1+513 | Released: 2025-06-05

c. Installing R Packages for the sessions

To save time during the training sessions, we recommend installing the following packages in advance. Internet access may be unavailable or unreliable during the training, which could prevent participants from installing the packages on-site and cause delays. The following code will install the main packages needed for the workshop.

```
install.packages(c("deSolve","dplyr"))
```

2. How to download and Install QGIS Software

QGIS is an open-source geographic information system that runs on Windows, macOS, Linux, Android and iOS. It provides a platform for users to create, edit and visualize geographical data. The app offers core plugins maintained by the QGIS.

To download QGIS, follow this link:

<https://qgis.org/download/>

Online (OSGeo4W) installer:

OSGeo4W Network Installer

This installer is the best way to keep QGIS up to date, run multiple versions on your system and servers to a minimum. [Learn more.](#)

Offline (Standalone) installers:

Long Term Version for Windows (3.40 LTR)

Latest Version for Windows (3.44)

Latest Version for Windows (3.44) with Qt6 (experimental)

Long Term Version for Windows (3.40) with Qt6 (experimental)

Choose the latest stable release. Typical options are:

- The *Standalone Installer* (recommended for most users)
- *Long Term Release (LTR)* for stability over time

To install:

- Run the Installer by locating the .exe file in your Downloads folder.
- Double-click to launch the installer.
- Follow the prompts: accept the license, choose the installation path, then click Install.
- After installation, you can open **QGIS Desktop** from your Start menu.

3. How to install KoboCollect app

- To install the app in your android device, follow these instructions
- Open Play Store (Android phone/device).
- In the search bar, type “KoBoCollect”.
- Tap on the KoBoCollect app (by KoboToolbox).
- Press Install and wait for download to complete.
- Once installed, please wait for more instructions to be shared during the training. So not open the app

4. How to join the morning and evening meeting sessions

All participants will be invited to join the **morning** and **evening** sessions via **two different** meeting links. The same meeting links will be used for **all the training** days, unless there is need to make adjustments. We have used your email address that was shared with us earlier to invite you to the meetings. In case you have a different email address that you wish to use for the meetings, please do let us know otherwise, you may not be admitted to the meeting with a different email address. In case you misplace or delete the invite, you can always use these links to join the meetings, but ensure you use the email address shared with us to avoid delays in joining the meetings.

Morning sessions: <https://meet.google.com/sft-atry-zkf>

Evening Sessions: <https://meet.google.com/wjb-rcxk-jnx>

During the training sessions via googlemeet, you can write questions on chat, or you can send a message on the chat section or raise your hand to be allowed to ask a question. **Finally, we request all participants to join the meeting at least 5 minutes early to enable us start the sessions on time.**

5. Evening Session: **Breakout rooms**

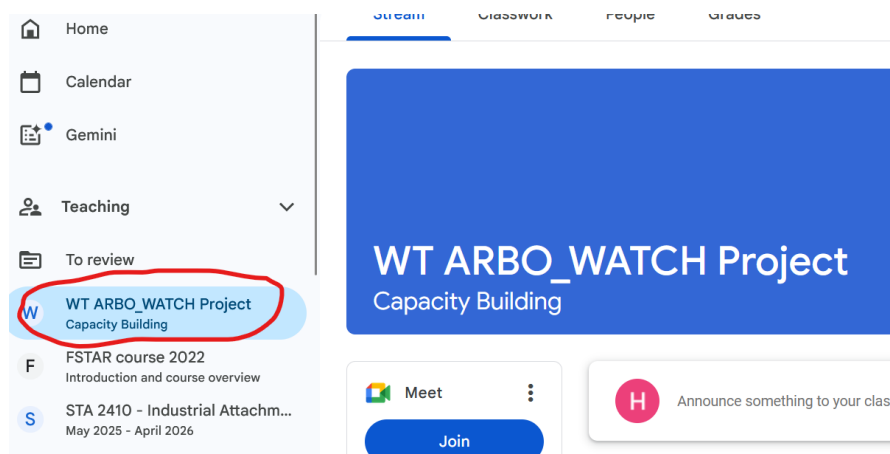
For the one-hour evening sessions, you will be assigned a group with one or two mentors. These breakout rooms will consist of 5-7 participants working with the mentors. These small groups are very essential to this training as it is impossible to take questions from 40 participants during the morning session. This is your opportunity to go through the assignments from the morning session and address any difficulties you had during the morning session. All participants will be invited to join the meeting via a meeting link. This will make it easier for you to join the meeting instantly, instead of waiting for someone to admit you to the meeting. However, in case you lose the link, you can still join the meeting using the following link: <https://meet.google.com/wjb-rcxk-jnx>

Please remember that there is no stupid question.

If you feel shy, remember you can message the facilitator using 'stream' and get some advice.

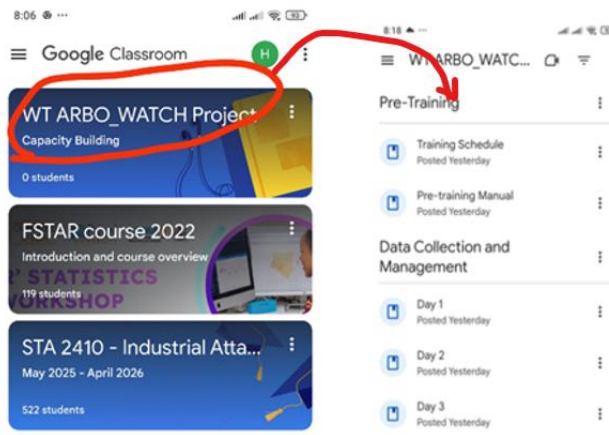
6. How to use the Google Classroom platform

We have set up a Google Classroom for this course to make it easier to share learning materials, post assignments, and upload session recordings. The figure below illustrates the Google Classroom interface. All participants will receive an invitation to join the classroom. Once registered, you will have access to course materials, receive announcements and updates, and also post messages on the platform. **The most reliable way to access the materials is directly through the platform, rather than through the email notifications you may receive about updates or changes.**



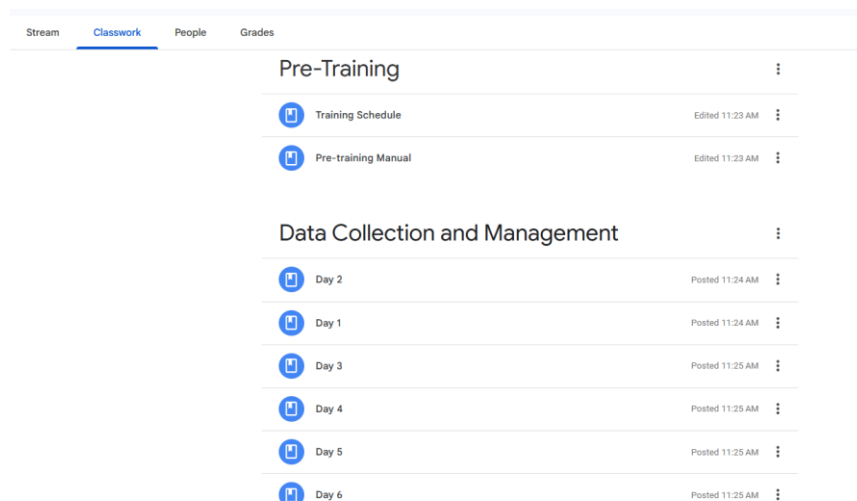
Optional access to google classroom

For quick and convenient access to the training materials, you may install the Google Classroom app on your phone or other Android device. To do this, open the Play Store, search for *Google Classroom*, and follow the installation instructions. After installation, log in using the email address you registered with for the classroom. You will then see a screen similar to the one shown below. Through the app, you can easily access all posted materials, including video recordings.



7. Google Classroom Materials

- After the morning session each day, you will be given some assignments. Your assignments will be in the **'classwork'** session of your course page.



- This is where all your assignments will be placed and where your facilitators will interact with you offline.
- To look up people, you can go to **'people'** located on the top of the page...

- You may also receive emails as a group from the facilitators and this you will also find at the top of the page.
- Any announcements posted will also be sent to you via email, and you can respond to the posts. Additionally, you can also ask questions by posting within the wall. Any of the facilitators or participants may respond to your post.
- All materials will also be available on the classwork page, which is linked to a Google Drive. We will also share a link to the Google Drive to ensure the material is available to all of you.